

Dynamic Capability Registry (DCR)

Overview

The Dynamic Capability Registry (DCR) is the central intelligence layer of ShivaAI Jarvis that enables a single AI assistant to dynamically identify domains, activate required skills, invoke tools, orchestrate workflows, and execute actions without relying on multiple hardcoded agents.

Instead of creating separate agents for Trading, Salesforce, Development, Research, Business Analysis, Automation, and Enterprise Operations, Jarvis maintains a registry of capabilities and dynamically assembles the required execution pipeline based on user intent.

Vision

Traditional Approach

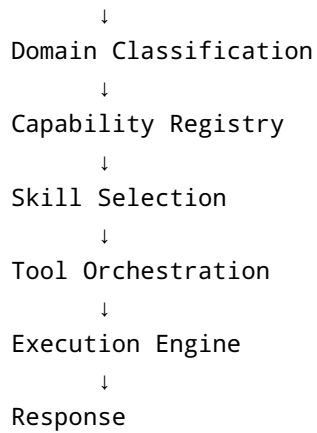
```
User
├─ Trading Agent
├─ Coding Agent
├─ Salesforce Agent
├─ Research Agent
├─ Business Analyst Agent
└─ Automation Agent
```

Problems:

- Agent duplication
 - Context switching
 - Complex maintenance
 - High infrastructure costs
 - Limited scalability
-

ShivaAI Approach

```
User
  ↓
Jarvis Core Brain
  ↓
Intent Detection
```



One Assistant

Unlimited Capabilities

Core Objectives

Dynamic Routing

Automatically determine:

- Domain
- Subdomain
- Required Skills
- Required Tools
- Memory Context
- Automation Possibility

Plug-and-Play Expansion

New capabilities can be added without modifying the AI core.

Example:

```
capability:
  name: Legal Assistant
  version: 1.0
```

Immediately available to Jarvis.

Autonomous Planning

Jarvis determines:

- What should be done
- In what sequence
- Which systems to use
- Whether execution is required

Registry Architecture

Dynamic Capability Registry

- ├ Domain Registry
- ├ Skill Registry
- ├ Tool Registry
- ├ Workflow Registry
- ├ Knowledge Registry
- ├ Connector Registry
- ├ Prompt Registry
- ├ Memory Registry
- └ Execution Registry

Domain Registry

Responsible for identifying the business domain.

Example Domains

Development

- React
- Angular
- NodeJS
- Python
- Salesforce
- DevOps

Trading

- Stocks
- Forex
- Crypto
- Futures
- Options

Enterprise

- Salesforce
- SAP
- Jira
- ServiceNow
- Confluence

Research

- Market Research
- Academic Research
- Competitive Analysis

Business Analysis

- User Stories
- Requirements
- Process Mapping
- Documentation

Skill Registry

Skills represent reusable intelligence blocks.

Example Skills

```
skills:

- document_analysis
- requirement_extraction
- story_generation
- coding
- architecture_design
- forecasting
- risk_analysis
- sentiment_analysis

```

- ui_generation
- workflow_creation

Capability Metadata

Each capability contains:

```
capability:
  id: trading_analysis

  name: Trading Analysis

  version: 2.0

  category: Trading

  skills:
    - sentiment_analysis
    - market_analysis
    - forecasting

  tools:
    - market_data_api
    - news_api

  confidence_threshold: 0.8

  priority: high
```

Tool Registry

Central catalog of all available tools.

Categories

AI Tools

- LLM
- Embeddings
- RAG

- Vision
- Voice

Development Tools

- GitHub
- GitLab
- Docker
- Kubernetes

Enterprise Tools

- Salesforce
- Jira
- Confluence
- SAP

Productivity

- Calendar
- Email
- Notion

Connector Registry

Stores all integrations.

Supported Integrations

CRM

- Salesforce
- HubSpot
- Zoho

Project Management

- Jira
- Azure DevOps
- ClickUp

Communication

- Slack
- Teams
- Discord

Cloud

- AWS
 - Azure
 - GCP
-

Knowledge Registry

Defines knowledge sources.

Sources

Internal

- Company Documents
- Project Documents
- SOPs
- Training Materials

External

- Web Search
- News APIs
- Research Databases

Personal

- User Notes
 - User History
 - User Preferences
-

Prompt Registry

Stores optimized prompts.

Categories

Coding

Code generation prompts.

Documentation

Documentation prompts.

Business Analysis

Story generation prompts.

Trading

Market analysis prompts.

Research

Research prompts.

Memory Registry

Controls memory retrieval.

Memory Types

Short-Term

Current conversation.

Session Memory

Current task.

Long-Term

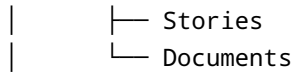
Historical interactions.

Knowledge Graph

Relationship storage.

Example:

```
graph TD
    User --> Project
    User --> Tasks
```



Workflow Registry

Stores executable workflows.

Example:

```
workflow:
  name: User Story Generation

  steps:
    - parse_document
    - extract_requirements
    - generate_stories
    - generate_acceptance_criteria
```

Execution Registry

Tracks execution capabilities.

Types

Read

Retrieve information.

Analyze

Perform reasoning.

Generate

Create output.

Execute

Perform actions.

Automate

Run workflows.

Intent Detection Engine

Determines user intent.

Example:

Input:

```
Create user stories from this PDF
```

Output:

```
{
  "intent": "story_generation",
  "domain": "business_analysis",
  "confidence": 0.97
}
```

Domain Classification Engine

Identifies primary domain.

Example:

```
Build Salesforce LWC Component
```

Result:

```
{
  "domain": "development",
  "subdomain": "salesforce",
  "confidence": 0.99
}
```

Capability Selection Engine

Selects required capabilities.

Example:

User:

Analyze uploaded BRD and create Jira stories.

Selected:

```
capabilities:
  - document_analysis
  - requirement_extraction
  - story_generation
  - jira_integration
```

Multi-Domain Orchestration

Example:

User Request:

Analyze BRD,
generate React UI,
create Salesforce objects,
prepare Jira stories.

Detected Domains:

Business Analysis
Development
Salesforce
Project Management

Activated Skills:

Document Parsing
Requirement Extraction
UI Design
Code Generation
Salesforce Mapping
Story Creation

Capability Dependency Graph

Document Analysis
↓
Requirement Extraction
↓
Story Generation
↓
UI Generation
↓
Code Generation
↓
Deployment

Capabilities automatically chain together.

Capability Marketplace

Future feature.

Organizations can install capabilities.

Examples:

- Salesforce Capability Pack
- Trading Capability Pack
- Healthcare Capability Pack
- Legal Capability Pack
- Cyber Security Capability Pack

No changes required in Jarvis Core.

Learning Engine

Jarvis continuously learns:

- Frequently used tools
- Preferred workflows
- User behavior
- Success rates

Capability ranking improves over time.

Observability

Metrics captured:

Performance

- Response Time
- Tool Latency
- Execution Time

Quality

- Success Rate
- User Satisfaction
- Accuracy

Usage

- Capability Usage
 - Domain Distribution
 - Workflow Frequency
-

Security

Every capability includes:

```
security:  
  authentication: required  
  authorization: RBAC
```

```
audit_logging: enabled
encryption: enabled
```

Enterprise Features

Governance

- Capability Approval
- Version Management
- Access Control

Compliance

- GDPR
- HIPAA
- SOC2

Audit Trail

Full execution history.

Future Roadmap

Phase 1

- Domain Detection
- Capability Registry
- Tool Registry
- Memory Registry

Phase 2

- Workflow Registry
- Capability Graph
- Auto Tool Selection

Phase 3

- Self-Learning Registry
- Capability Marketplace
- Dynamic Skill Creation

Phase 4

- Autonomous Planning
 - Autonomous Execution
 - Self-Optimizing Intelligence Layer
-

Final Vision

Dynamic Capability Registry transforms ShivaAI Jarvis into a Universal AI Operating System where a single AI can dynamically understand any domain, activate the correct skills, orchestrate tools, execute workflows, and continuously improve without requiring separate agents.

The registry becomes the foundation for scalability, extensibility, enterprise adoption, and future autonomous intelligence.